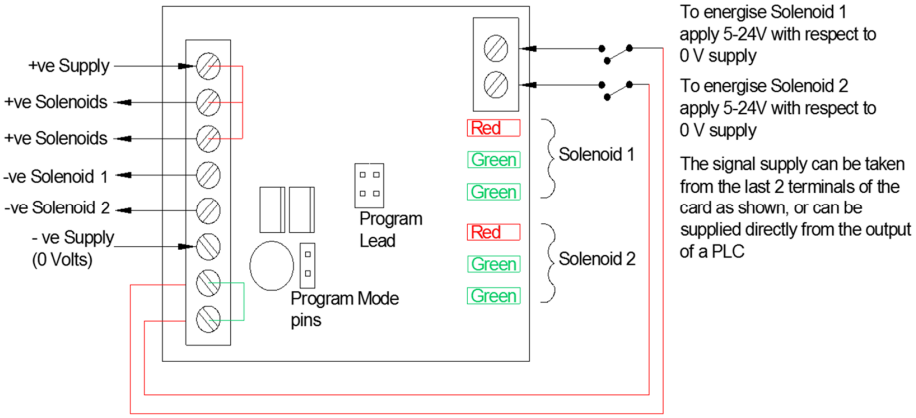
Actuation Cards

Selection data & performance | Manufacturer technical reference.

The directional control valves are supplied with a 12V DC coil*. The PCB will control the duration of supply to ensure the maximum temperature of the coil is not exceeded. The solenoid should not be switched on/off continuously with a duty cycle shorter than 1 second.

Power supply

A power supply capable of supplying a regulated 24V DC (supply) at 5 amps is required to the actuation card at all times.
The schematic below shows the signal supply can be derived from the last 2 terminals of the board; alternatively the signal can be supplied directly from a PLC (6mA if the output from the PLC is 24V or 1mA if the output is 5V).



With the connections made as shown, by supplying a 5-24V signal to the input a sequence will commence. The solenoid will be supplied with an initial 24V supply for 100ms. The average voltage the solenoid will then see is 15V for a further 2 seconds. A further reduction to 10V is then made to reduce the temperature generated in the solenoid under hold conditions; this will also reduce the power consumption of the system.
The sequence is illustrated by the LED's; the red is the holding condition.

Please note: once energised, the card will remain in the holding condition with a signal voltage of only 1.3V, therefore care should be taken when routing the signal cables to avoid interference which may induce voltages high enough to trigger the signal.

Temperature

The rated operating range for this module is –40°C to 50°C ambient.

Mounting

The board can be supplied with optional DIN rail mounting as shown below.

Din Rail Mounting





Match card type, control signal and electrical supply to the selected valve actuator. Specify enclosure, wiring, interlocks and commissioning requirements. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

Actuation Cards V.01.25.pdf, p. 1. Manufacturer labels and dimensions retained.

Actuation Cards

Dimensions, connections & configuration | Manufacturer technical reference.

ORDERING CODES	24V	48V**	DIMENSIONS
Actuation Card Only	004-010	004-013	H 38mm L 78mm W 42mm
1 Card including Din Rail Mounting	004-104	004-114	H50mm L 80mm W 45mm
2 Cards including Din Rail Mounting	004-105	004-115	H 50mm L 80mm W 90mm
3 Cards including Din Rail Mounting	004-106	004-116	H 50mm L 80mm W 136mm
Mounting Only (1 Card)	004-101		H 50mm L 80mm W 45mm
Mounting Only (2 Cards)	004-102		H 50mm L 80mm W 90mm
Mounting Only (3 Card)	004-103		H 50mm L 80mm W 136mm

*Specially programmed versions available for 28v automotive applications
**48v coil version requires 24v and 48v 2.5A PSU

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Actuation Cards V.01.25.pdf, p. 2. Manufacturer labels and dimensions retained.